

SIH26126 — Mission Report

Recorded run: run_final · goal UTM 44N E 403562.52 N 1450024.12 (map 56.00, 34.00) · no GNSS in the loop

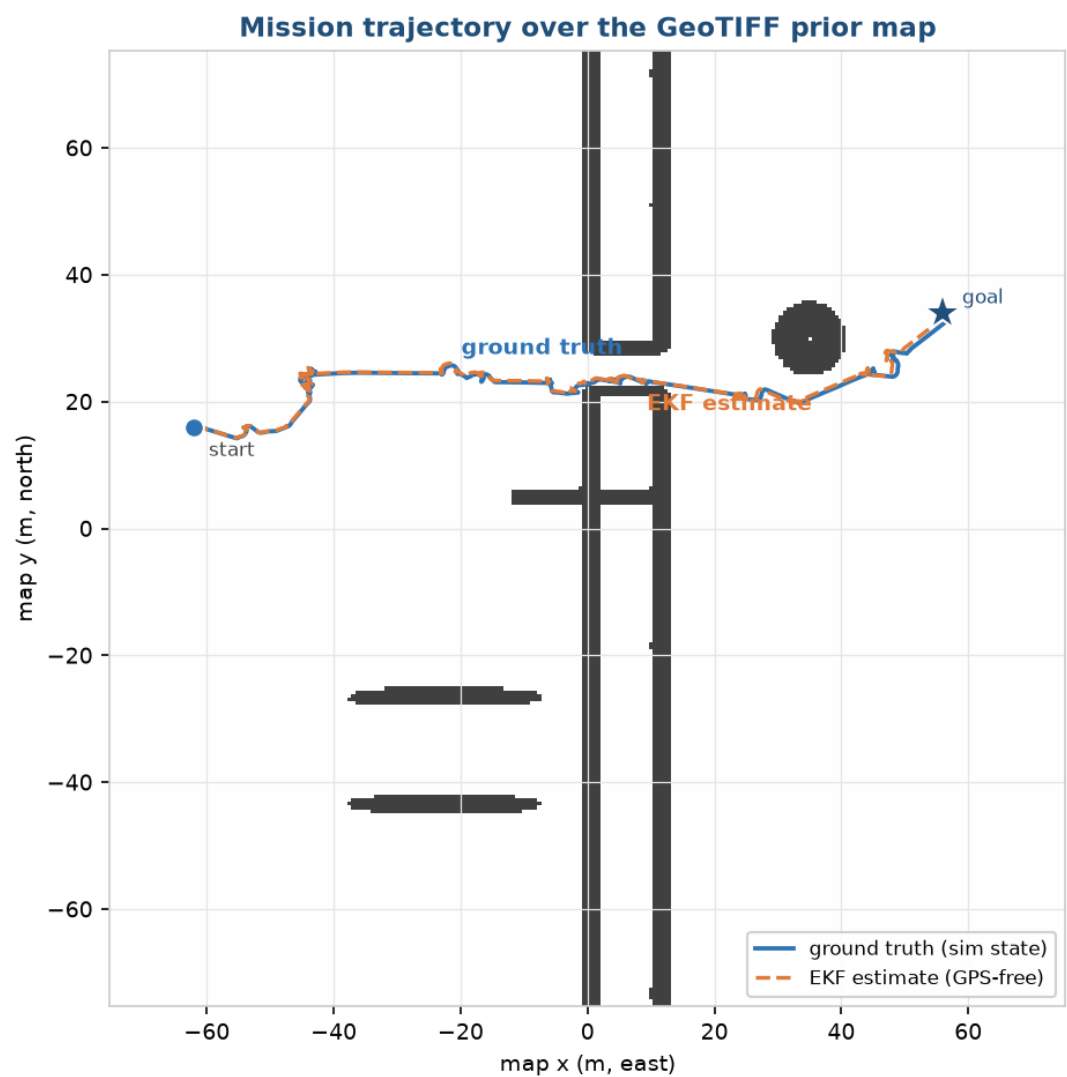
MISSION SUCCESSFUL — GOAL REACHED

1. Result summary

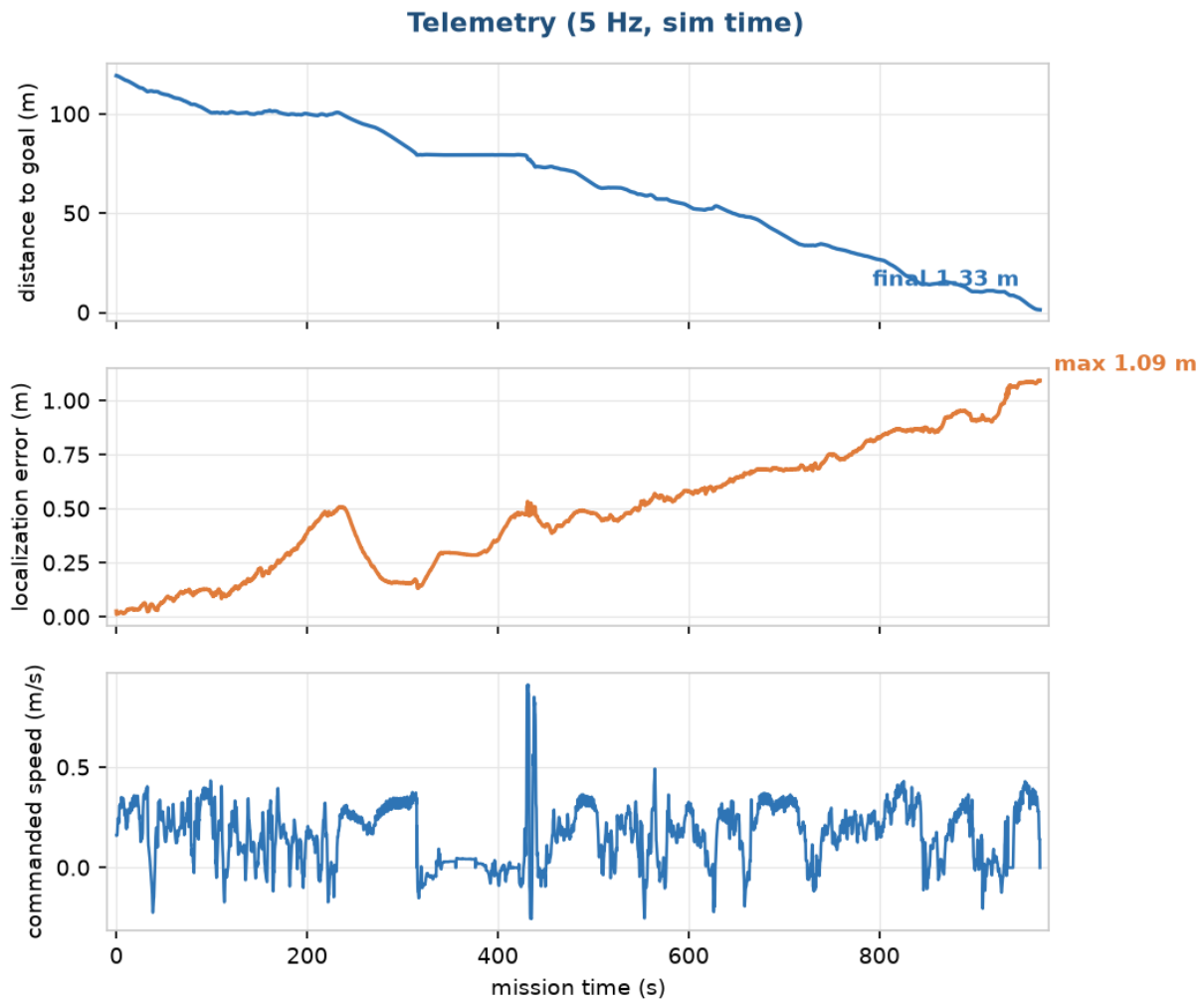
Quantity	Value	Source
Initial distance to goal	119.34 m	ground truth at first log row
Final distance to goal	1.33 m	ground truth
Localization error at goal	1.09 m	ground truth – EKF — the GPS-free claim
Mean / max localization error	0.49 / 1.09 m	whole mission
Path length driven	189.05 m	integrated ground truth
Mission duration	968 s sim time (16.1 min)	telemetry log
Time in motion	759 s	cmd_v > 0.05 m/s
Average moving speed	0.25 m/s	path / moving time
Recovery manoeuvres	7	Nav2 feedback
Nav2 final status	4 (SUCCEEDED)	NavigateToPose result

Ground truth comes from the simulator state and is used for scoring only — nothing in the navigation stack subscribes to it. The EKF position is what the rover actually believed and navigated on, computed from body-frame velocities and IMU heading alone.

2. Trajectory

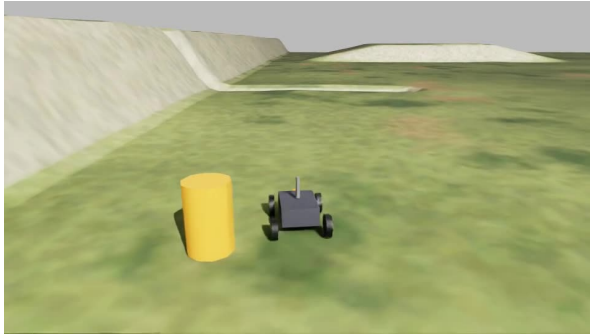


3. Telemetry

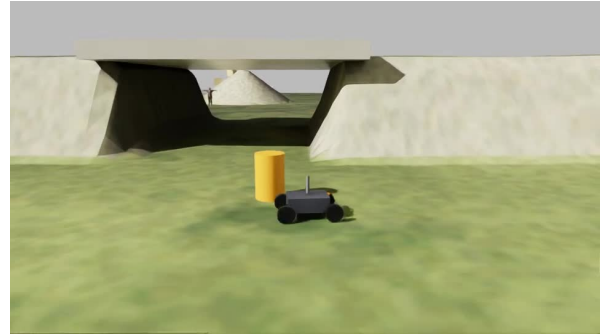


Flat stretches in distance-to-goal with nonzero speed are the tunnel and detours around discovered obstacles; drops to zero speed mark recovery manoeuvres.

4. Camera perspectives (from the recordings)



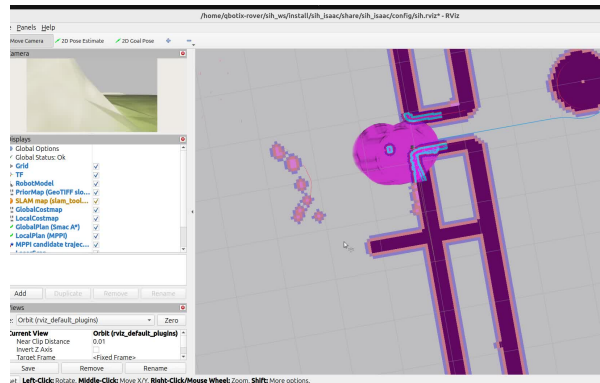
behind the rover



side angle



top-down



RViz (the robot's belief)

5. Recording inventory

File	Duration	Size
isaac_behind.mp4	990 s	36.7 MB
isaac_side.mp4	990 s	39.2 MB
isaac_top.mp4	990 s	19.3 MB
rviz.mp4	926 s	23.0 MB

All files are in `/home/qbotix-rover/sih_ws/mission_recordings/run_final`. The three `isaac_*.mp4` views are rendered in-simulation at 15 fps of sim time (behind / side follow cameras and a heading-stabilised top-down view); `rviz.mp4` is a screen capture of the navigation stack's live belief — costmaps, plans, lidar and TF.